NEEL JACOB JOGY

Junior Python Developer · Backend Engineer · MLOps

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PROFESSIONAL SUMMARY

Software Engineering graduate (2:1, University of Greenwich) with hands-on experience building Python backends, REST APIs, and ML-powered systems. Built and deployed a hybrid ArcFace + CNN face recognition API using FastAPI and TensorFlow trained on VGGFace2 — achieving 100% softmax confidence on test images. Comfortable across the full ML pipeline from model training and evaluation through to API deployment and data handling. Seeking a Junior Python/Backend, MLOps, or Graduate Software Engineer role.

TECHNICAL SKILLS

Languages: Python (primary), SQL, Java, JavaScript (basic)
Backend & APIs: FastAPI, Flask, REST APIs, Uvicorn, Python-Multipart, Supabase Auth & RLS
ML & AI: TensorFlow, Keras, InsightFace (ArcFace), OpenCV, scikit-learn, NumPy, CNNs, cosine similarity
Data & Pipelines: Pandas, NumPy, data preprocessing, model inference pipelines, gdown
Databases: PostgreSQL, SQLite, Supabase
Tools: Git, GitHub, Postman, VS Code, Jupyter Notebooks

PROJECTS

Hybrid Face Recognition API — Python / FastAPI / TensorFlow / InsightFace ↗ GitHub

- End-to-end ML pipeline: VGGFace2 dataset → ArcFace + CNN dual-branch training → REST API with automatic model download on startup
- Achieved 100% softmax confidence and 74.7% cosine similarity on test images using threshold-based dual decision logic
- Exposed via FastAPI with Swagger docs; built Tkinter GUI and Flask web app for real-time webcam, video, and image recognition
- Integrated Wikipedia API and voice feedback; Stack: Python, TensorFlow, Keras, FastAPI, InsightFace, OpenCV, scikit-learn, gdown

Open-To-Work Candidate Tracker — FastAPI / Supabase / PostgreSQL ↗ GitHub

- Multi-user backend API with Supabase Auth, row-level security (RLS), and RESTful CRUD endpoints for candidate management
- Designed normalised PostgreSQL schema with access control policies; Stack: Python, FastAPI, Supabase, PostgreSQL

London Underground Route Planner — Python / Algorithms

- Route optimisation using Dijkstra's algorithm with time-based and station-count routing across the TfL network
- Built data visualisation tools to analyse network efficiency and journey times across stations

Distributed Network System — Java / Client-Server

- Java-based distributed system with group communication, fault-tolerant messaging, and client-server architecture
- Conducted unit testing using JUnit to validate message delivery and system reliability

EDUCATION

University of Greenwich · *BEng (Hons) Software Engineering — 2:1* · 2022 – 2025

- Relevant modules: Software Architecture, Distributed Systems, Database Systems, Algorithms & Data Structures

WORK EXPERIENCE

Primark · *Retail Assistant* · March 2023 – Present

- Consistent delivery in a high-volume environment while studying full-time — strong communication, teamwork and time management under pressure

Tesco · *Customer Assistant* · November 2022 – February 2023

- Provided customer support and resolved issues efficiently in a collaborative team environment